Tricky Topic

Simultaneous equations

Simultaneous equations have two unknowns. You need to find the values for the two unknowns which make **both** equations true.

Algebraic solution

1. Number each equation.
2. If necessary, multiply the equations so that the coefficients of one unknown are the same.
3. Add or subtract the equations to **eliminate** that unknown.
4. Once one unknown is found use substitution to find the other.
5. Check the answer by substituting both values into the original equations.

$3x + y = 20$ ①
$x + 4y = 14$ ②
$12x + 4y = 80$ ① × 4
$-(x + 4y = 14)$ −②

$11x = 66$
$x = 6$

Substitute $x = 6$ into ①:
$3(6) + y = 20$
$18 + y = 20$
$y = 2$

Solution is $x = 6$, $y = 2$.
Check: $x + 4y = 6 + 4(2) = 14$ ✓

Worked example

Solve the simultaneous equations
$6x + 2y = -3$ ①
$4x - 3y = 11$ ② **(4 marks)**

$18x + 6y = -9$ ① × 3
$+ 8x - 6y = 22$ ② × 2

$26x = 13$
$x = \frac{1}{2}$

Substitute $x = \frac{1}{2}$ into ①:
$6\left(\frac{1}{2}\right) + 2y = -3$
$3 + 2y = -3$
$2y = -6$
$y = -3$

Solution is $x = \frac{1}{2}$, $y = -3$

Multiply both equations by a whole number to make the coefficients of y the same.

Check it!
Always use the equation you **didn't** substitute into to check your answer:
$4x - 3y = 4\left(\frac{1}{2}\right) - 3(-3) = 2 + 9 = 11$ ✓

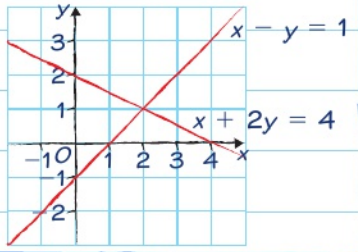
Easier eliminations

You can save time by choosing the right unknown to eliminate. Look for one of these:

1. If an unknown appears **on its own** in one equation you only need to multiply one equation to eliminate that unknown.
2. If an unknown has **different signs** in the two equations you can eliminate by **adding**.

Graphical solution

You can solve these simultaneous equations by drawing a graph.
$x - y = 1$ $x + 2y = 4$
The coordinates of the point of intersection give the solution to the simultaneous equations.
The solution is $x = 2$, $y = 1$



Now try this

1 Solve the simultaneous equations
$3x - 2y = 12$
$x + 4y = 11$ **(3 marks)**

Draw a coordinate grid from −3 to 5 in both directions.

2 By drawing two suitable straight lines on a coordinate grid, solve these simultaneous equations.
$3y + 2x = 6$
$y = 2x - 2$ **(4 marks)**

Worked solution video

Tricky Topic

Rearranging formulae

Most formulae have one letter on its own on one side of the formula. This letter is called the **subject** of the formula.

$P = 2l + 2w$ P is the subject.

$A = \frac{1}{2}bh$ A is the subject.

You can use algebra to change the subject of a formula. This is like solving an equation.

If you need to find a missing value which is not the subject in a formula:

1. Substitute any values you know into the formula.

For a reminder about substitution have a look at page 25.

2. Solve the equation to find the missing value.

Worked example

Alicia works h hours of normal time and v hours of overtime each week. She is paid £P. She uses this formula to work out her pay:

$P = 8h + 12v$

Last week Alicia worked 32 hours of normal time and was paid £328. How many hours of overtime did she work? **(3 marks)**

$328 = 8 \times 32 + 12v$

$328 = 256 + 12v$ (-256)

$72 = 12v$ $(\div 12)$

$6 = v$

Alicia worked 6 hours of overtime.

To make p the subject of this formula you have to do the same thing to **both sides** of the formula until you have p on its own on one side.

$N = 2p + 3q^2$ $(-3q^2)$ — Subtract any terms you don't need.

$N - 3q^2 = 2p$ $(\div 2)$ — $2p$ means $2 \times p$ so divide both sides by 2 to get p on its own.

You have to divide **everything** on this side by 2.

$\frac{N - 3q^2}{2} = p$

Worked example

Rearrange this formula to make Q the subject: **(2 marks)**

$R = 1 - 5Q$

$R = 1 - 5Q$ $(+5Q)$

$R + 5Q = 1$ $(-R)$

$5Q = 1 - R$ $(\div 5)$

$Q = \frac{1 - R}{5}$

You want the term containing Q to be positive. Start by adding $5Q$ to both sides.

Worked example

Rearrange this formula to make h the subject: **(2 marks)**

$A = \frac{1}{2}bh$

$A = \frac{1}{2}bh$ $(\times 2)$

$2A = bh$ $(\div b)$

$\frac{2A}{b} = h$

You need the new subject on its own on one side of the formula. It doesn't matter which side.

Now try this

Worked solution video

1 (a) Rearrange this formula to make Q the subject:

$W = 7Q - 100$ **(2 marks)**

(b) Rearrange this formula to make V the subject:

$d = \frac{m}{V}$ **(2 marks)**

2 Rearrange this formula to make n the subject:

$p = 2(m - n)$ **(2 marks)**

Start by expanding the brackets.

For a reminder about expanding brackets look at page 28.

Using algebra

You can sometimes use the information given in a question to write an equation.
You might need to choose a letter to represent an unknown quantity you are trying to find.

Worked example

Here are three number cards.



Each card has a whole number written on the back.
The number on card **B** is twice the number on card **A**.
The number on card **C** is five more than the number on card **A**.
The sum of the numbers on all three cards is 37.
Work out the number on each card. **(4 marks)**

$B = 2A$ $\quad C = A + 5$

$A + B + C = 37$

$A + 2A + (A + 5) = 37$

$4A + 5 = 37 \quad (-5)$

$4A = 32 \quad (\div 4)$

$A = 8$

$B = 2 \times 8 = 16 \quad C = 8 + 5 = 13$

Problem solved! You could try guessing the numbers on the cards, but it might take a long time. You can solve this problem quickly by writing your own equation.

- Use A to represent the number on card **A**.
- Write expressions for the numbers on cards **B** and **C** in terms of A.
- Add together these three expressions and set the total equal to 37. Solve this equation to find A.

Once you have calculated A, remember to work out the values of B and C as well.

Check it!

$8 + 16 + 13 = 37$ ✓

True or false?

You might need to **substitute** values into an algebraic statement to show that it is **false**. You only need to find one **counter-example** to show that a statement is false.

Try some different values of n and write down all your working.
You should know the prime numbers up to 20.

Worked example

n is a whole number.
Supraj says that $2n^2 + 3$ is always a prime number.
Give an example to show that he is wrong. **(2 marks)**

$n = 1$: $2n^2 + 3 = 2 \times 1 + 3 = 5$ Prime

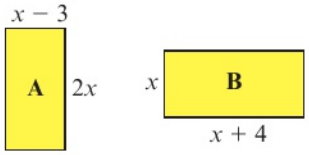
$n = 2$: $2n^2 + 3 = 2 \times 4 + 3 = 11$ Prime

$n = 3$: $2n^2 + 3 = 2 \times 9 + 3 = 21 = 3 \times 7$ Not prime

Worked solution video

Now try this

1 These two rectangles have the same perimeter.



In the diagram, all of the measurements are in metres. Work out the width and height of rectangle **A**. Show all of your working. **(4 marks)**

Start by writing expressions for the perimeter of each rectangle. The perimeter of rectangle A is:
$(x - 3) + 2x + (x - 3) + 2x$
Remember to answer the question at the end.

2 Jill says that the expression $n^2 + 4n + 3$, where n is an integer, is never a multiple of 7.
Give an example to show that Jill is wrong. **(2 marks)**

Tricky Topic

Identities and proof

An **identity** is something that is **always** true. The right-hand side will always be exactly equal to the left-hand side, no matter what values you substitute.
You use the symbol '≡' to represent an identity. You might need to **show** that an identity is true in your exam.

Golden rule

An identity is not like an equation. Do not solve it using the balance method.
Manipulate each side separately. ✓
Apply the same operation to both sides. ✗

Worked example

Show that $(n - 1)^2 + (n + 1)^2 \equiv 2(n^2 + 1)$ **(2 marks)**

$(n - 1)^2 + (n + 1)^2$
$\equiv (n - 1)(n - 1) + (n + 1)(n + 1)$
$\equiv n^2 - 2n + 1 + n^2 + 2n + 1$
$\equiv 2n^2 + 2$
$\equiv 2(n^2 + 1)$

If you need to **show** that an identity is true in an exam you should show every line of your working.
Start with the expression on the left-hand side. Use multiplying out, simplifying and factorising to work towards the expression on the right-hand side.

Proving results with algebra

You can use algebra to show that numbers are **odd**, **even**, or **multiples** of another number. You can use n to represent any integer. You might need to **rearrange** or **simplify** algebraic expressions, and you might have to write a **conclusion**. Use the table on the right to help you.

Number fact	Written using algebra
even number	$2n$
odd number	$2n + 1$ or $2n - 1$
multiple of 3	$3n$
consecutive numbers	n, $n + 1$, $n + 2$, ...
consecutive even numbers	$2n$, $2n + 2$, $2n + 4$, ...
consecutive odd numbers	$2n + 1$, $2n + 3$, $2n + 5$, ...
consecutive square numbers	n^2, $(n + 1)^2$, $(n + 2)^2$, ...

Multiple or not?

If n is an integer then $3n$ is a **multiple of 3**. This means that $3n + 1$ and $3n + 2$ are **not** multiples of 3.

Problem solved! You can show that this is true for **any** three consecutive integers using algebra.

1. Write the first integer as n and the next two integers as $n + 1$ and $n + 2$.
2. Write an expression for the sum of your three integers. Simplify your expression.
3. Factorise your expression then write a conclusion.

Worked example

Show that the sum of any three consecutive integers is always a multiple of 3. **(3 marks)**

n, $n + 1$ and $n + 2$ represent any three consecutive integers.
$n + (n + 1) + (n + 2) = 3n + 3$
$= 3(n + 1)$
$n + 1$ is an integer, so $3(n + 1)$ is a multiple of 3.

Now try this

1 Show that $(a + b)(a - b) \equiv a^2 - b^2$ **(2 marks)**

2 Given that $4(x - n) = 3x - 1$ where n is an integer, show that x is an odd number. **(3 marks)**